



Overview All Automotive Simulation Tools

Automotive simulation tools are software applications/engines used to model, simulate and resultingly visualize or test various aspects of automotive design. Some tools are focused on vehicle dynamics, analyzing how a vehicle behaves in different conditions and how it responds to different inputs (Engineering, Driving Simulators & Game Development). Other tools are designed to help simulate the performance of individual vehicle components, advanced driver assistance systems (ADAS) or autonomous driving (AD) functions.

Compare & Search – Automotive Simulation Tools Table

Find simulation tools fulfilling your engineering or game development needs.
(Hover/click on rows to expand)

Tool	Features	Use Case	Forma
		All	All
A/B Street	Traffic, Maps, OpenStreetMap	Engineer.	OSS (C
Abaqus	CAE, FEA	Engineer.	Deskt
Actran	CAE, NVH, Acoustics	Engineer.	Deskt
AcuSolve	CAE, CFD	Engineer.	Deskt
Adams	CAE, HIL	Engineer.	Deskt
Adams Car	CAE, Dynamics, HIL, Suspension, Steering,...	Engineer.	Deskt
Adams Flex	CAE, FEA	Engineer.	Deskt
Aimsun	Traffic, Maps	Engineer.	Deskt
AirSim	ADAS, AD	Engineer.	OSS (C
ALPHA DRIVE	Cloud based simulator, CARLA, UNITY, Un...	Engineer.	Deskt
Altair Activate	CAE, MBD, Digital Twin	Engineer.	Deskt
Altair Flux	CAE, Magnetics, Electric	Engineer.	Deskt

Tool	Features	Use Case	Forma
Altair Inspire	CAE	Engineer.	Desktop
Altair MotionSolve	Multibody System Simulation, Discrete El...	Engineer.	Desktop
ANSA	CAE	Engineer.	Desktop
ANSYS	CAE, Mechanical, CFD, Thermal, Acoustics...	Engineer.	Desktop
Ansys CFX Industry-Leading C...	CFD, Flow, Turbomachinery	Engineer.	Desktop
Ansys Fluent	CAE, Fluids	Engineer.	Desktop
Ansys LS-DYNA Crash Simulat...	CAE, Drop tests, Impact and Penetration, ...	Engineer.	Desktop
Ansys Mechanical Structural ...	CAE, FEA, Structures	Engineer.	Desktop
Ansys Motion	Multibody Dynamics	Engineer.	Desktop
Ansys Motor-CAD Electromec...	CAE, Motor, Magnetics, Thermal	Engineer.	Desktop
Ansys Rocky Particle Dynamic...	CAE, Particles, Flow	Engineer.	Desktop
Ansys Sherlock Electronics R...	CAE, ECAD, FEA, CFD, Electric	Engineer.	Desktop
Ansys Thermal Desktop	CAE, Heat Transfer, Fluids, Thermal	Engineer.	Desktop
AnyLogic	Traffic, Maps	Engineer.	Desktop
Apollo (by Baidu)	ADAS, AD	Engineer.	OSS (C
AppGameKit	Game Development	Game	Desktop
ArduPilot	Autopilot, Controls	Engineer./...	OSS (C
AutoForm	CAE, Sheet Metal	Engineer.	Desktop
Automated Driving Toolbox (...)	ADAS, AD, Maps, OpenDRIVE, SIL, HIL, Si...	Engineer.	Desktop
Autoware	ADAS, AD, ROS	Engineer.	OSS (C
AVL CAMEO	CAE, SIL, ADAS, Battery, Fuel Cell, Hybrid	Engineer.	Desktop
AVL Cruise M	CAE, Engine, Fow, Driveline, Electrics, Hyd...	Engineer.	Desktop
AVL Excite	CAE, MBD, Multi-Body Dynamics, Structur...	Engineer.	Desktop
AVL Fire M	CAE, Engine, Emissions	Engineer.	Desktop
AVL Powertrain Integration	Hybrid EVs, Plug-in hybrid EVs, Battery EV...	Engineer.	Desktop
AVL PUMA	HIL, Powertrain	Engineer.	Desktop
AVL Scenario Simulator	Vehicle Dynamics, OpenSCENARIO, Scena...	Engineer.	Desktop
AVL SCENIUS Scenario Designer	CAE, Scenario-Based Testing, OpenSCEN...	Engineer.	Desktop
AVL SiL Suite	MBD, SIL, HIL	Engineer.	Desktop
AVL ThreatGet	MBD, Threat Analysis, Risk Assessment, C...	Engineer.	Desktop
AVL VSM (Vehicle Simulation ...)	HIL, Real-time, Comfort Calibration, Nonli...	Engineer.	Desktop
AVL-DRIVE	Vehicle Attribute Assessment	Engineer.	Desktop
AWSIM & AWSIM Labs	Autoware, AD, ADAS, Unity, Scenario-Bas...	Engineer.	Desktop

Tool	Features	Use Case	Forma
BeamNG.tech	MIL, ADAS, Sensors, Soft-Body Physics, Dr...	Engineer.	OSS (C
CAESES	CAE, CAD, CFD	Engineer.	Deskt
CANalyzer/CANape/CANoe	ECU Calibration & diagnostics, Restbus Si...	Engineer.	Deskt
CARLA	Traffic, Maps, Dynamics, AD, Autoware, R...	Engineer.	OSS (C
CarMaker Realtime	Dynamics, Scenario-Based Testing, HIL, R...	Engineer.	Deskt
CarMaker/TruckMaker/BikeMa...	Dynamics, Scenario-Based Testing, MIL, SI...	Engineer.	Deskt
CARNET	Driving Simulator, Dynamics, HIL	Engineer.	Deskt
CarSim/TruckSim/BikeSim	Dynamics, Scenario-Based Testing, MIL, SI...	Engineer.	Deskt
CATIA	CAE, CAD, CFD, FEA, Fluids, Electrical, Str...	Engineer.	Deskt
ChassisSim	Track Simulator, Driving Simulator, Dyna...	Engineer.	Deskt
CityFlow	Traffic, Maps	Engineer.	OSS (C
Cocos2d	C++, Lua, and JavaScript, HMI, 3D Car Mo...	Game	Deskt
CommonRoad	Motion planning benchmarks, Scenario da...	Engineer.	OSS, E
Construct	2D Game Engine, Javascript	Game	Deskt
CONVERGE CFD	CFD	Engineer.	Deskt
CosiMate	Co-simulation, FMI, Modelica, C, C++, Java...	Engineer.	Deskt
Cradle CFD	CFD, Fluids	Engineer.	Deskt
CryEngine	Game Development, 3D	Game	OSS (C
DARTS	Kinematic & dynamic applications in robo...	Engineer.	OSS (C
Defold	Game Development, 2D, 3D, Free	Game	OSS, E
DIAdem	Data Analysis & Visualization	Engineer.	Deskt
DriveSim	AD, ADAS, Sensors, HIL, GPU, CUDA	Engineer.	SDK
dSPACE Automotive Simulatio...	HIL, Dynamics, Combustion Engine, Electr...	Engineer.	Deskt
Dymola	Modelica, FMI, MBD, Catia, HIL	Engineer.	Deskt
DYNA4 (Vector).	Virtual Test Driving, Vector, Real-Time, Dy...	Engineer.	Deskt
Easy5	CAE, Control systems, Hydraulics, Pneum...	Engineer.	Deskt
Engine Analyzer	Engine	Engineer.	Deskt
esmini	OSI, OpenDrive, OpenScenario, Unity, Tra...	Engineer.	OSS (C
FASTSim (Future Automotive S...	Engine, Powertrain, Electrical	Engineer.	Excel,
FEMFAT	Fatigue Analysis, Lightweight Design	Engineer.	Deskt
Fidelity CFD Software (Cadence)	CAE, CFD, Acoustics	Engineer.	Deskt
Flow (Traffic Control Benchma...	Traffic, Maps, ML, SUMO, Aimsun	Engineer.	OSS (C
FLOW-3D	CAE, CFD, Flow, Liquids	Engineer.	Deskt

Tool	Features	Use Case	Forma
Functional Mock-up Interface (...)	FMI, FMU, Co-Simulation	Engineer.	OSS (C
GameMaker Studio	Game Engine, 2D, 3D	Game	Deskt
Gazebo	Robotics, ROS, 3D visualizer, Sensors, Phy...	Engineer.	Deskt
Godot	Game Engine, 2D, 3D	Game	Deskt
GT-Power (Gamma Technologi...	Engine	Engineer.	Deskt
GT-SUITE	CAE, Integrated Multi-physics Systems Si...	Engineer.	Deskt
HaxeFlixel	Game Engine, 2D	Game	Deskt
HEEDS	CAE, Optimization	Engineer.	Deskt
HiL Sim	HiL, AD, ADAS	Engineer.	Deskt
HyperMesh	CAE, FEM	Engineer.	Deskt
HyperStudy	CAE, Design Exploration and Optimization	Engineer.	Deskt
HyperWorks	CAE, Structures, Motion, Fluids, Thermal, ...	Engineer.	Deskt
ICFD++	CAE, CFD, Fluids	Engineer.	Deskt
Interactive Visualization Lab	Game Development, AR/VR, Digital Twin, ...	Game/Eng...	Deskt
Isight	CAE, Optimization	Engineer.	Deskt
IVEX OpenScenario XML Expo...	OpenScenario, OpenDRIVE, Scenario-Bas...	Engineer.	Deskt
IVEX Scenario Mining Tool	AD, ADAS, Scenario-Based, Scenario Mining	Engineer.	Deskt
JMAC	Motor, Thermal, Electromagnetics	Engineer.	Deskt
KISSsoft	Motor, Gears, Thermal, Electromagnetics	Engineer.	Deskt
KULI	Thermal, Vehicle heating	Engineer.	Deskt
LibGDX	Game Engine, 2D, 3D	Game	OSS (C
MAGMASOFT	Metal Casting Simulation	Engineer.	Deskt
MapleSim	Modelica, Multidomain, MBD, Dynamics, F...	Engineer.	Deskt
MATLAB/Simulink	Simulink, MBD, Real-Time, SIL, Physics, Sig...	Engineer.	Deskt
MATSim	Agent-Based Traffic Simulation	Engineer.	OSS (C
MAZE	Road Generation, OpenDRIVE, Maps	Engineer.	Deskt
MBDyn	Multidomain, Mechanical, aeroservoelasti...	Engineer.	OSS (C
Modelica	Multidomain, Physics, Math	Engineer.	Langu
Modelon Impact	Multidomain, Physics, Math, Modelica, CL...	Engineer.	Deskt
Modelon Library Suite	Multidomain, Physics, Math, Modelica, Ve...	Engineer.	Deskt
MonoGame	Game Development, 2D, 3D, Desktop, Mo...	Game	OSS (C
Morse Simulator	Robotics, ROS, YARP, 3D visualizer, Senso...	Engineer.	Deskt
Nastran	CAE, FEA	Engineer.	Deskt

Tool	Features	Use Case	Forma
nCode DesignLife	CAE, Fatigue analysis	Engineer.	Desktop
NetLogo Traffic Simulation	Traffic	Engineer.	Desktop
NextSim	CFD, Aeronautical	Engineer.	Desktop
O3DE	Game Engine, 2D, 3D	Engineer./...	OSS (C)
OpenFOAM	Aerodynamics	Engineer.	OSS (C)
OpenModelica	Multidomain, Physics, Math, Modelica	Engineer.	Desktop
OpenTraffic	Traffic, OpenStreetMap	Engineer.	OSS (C)
OpenTrafficSim	Traffic, OpenStreetMap	Engineer.	OSS (C)
OptimumKinematics	Kinematics, Suspension	Engineer.	Desktop
OptimumLap	Vehicle Dynamics, Lap Time Optimization	Engineer.	Desktop
OptimumTire2	Tires	Engineer.	Desktop
Optistruct	CAE, FEA, FEM, Nastran	Engineer.	Desktop
OSM2VISSIM	Traffic, VISSIM, Openstreetmap	Engineer.	OSS (C)
OSMWebWizard	Traffic, SUMO, Openstreetmap	Engineer.	OSS (C)
Panda3D	Game Engine, 3D, C++, Python	Game	Desktop
Permas	CAE, Finite Element, Thermal, Hear, Acou...	Engineer.	Desktop
PhysX	Physics Simulation, GPU	Engineer./...	OSS (C)
Plexe	Traffic, SUMO, Veins, Scenario-Based, Veh...	Engineer.	OSS (C)
PowerACOUSTICS	CAE, NVH, Vibration, Acoustics	Engineer.	Desktop
PowerFLOW	CAE, CFD, Flow, Liquids	Engineer.	Desktop
PowerTHERM	CAE, Thermal, Cooling	Engineer.	Desktop
PreScan	ADAS, AD, Physics-based Sensor Simulati...	Engineer.	Desktop
PTV Vissim	Traffic, Maps, 3D Visualization, Pedestria...	Engineer.	Desktop
RecurDyn	CAE, Multi-Body Dynamics	Engineer.	Desktop
rEpro	Vehicle Dynamics, Driving Simulation, AD,...	Engineer.	Desktop
Ricardo IGNITE	Physics-based, Energy, Powertrain, Fuel c...	Engineer.	Desktop
Ricardo VECTIS	CFD, Combustion Engines	Engineer.	Desktop
Ricardo Wave & WAVE-RT	CFD, Combustion Engines, SIL, HIL, MIL, ...	Engineer.	Desktop
RoadDesign	Maps, Road	Engineer.	Desktop
RoadRunner (Mathworks).	Maps, Road, OpenDrive, Traffic Signals, A...	Engineer.	Desktop
Romax	CAE, NVH, Electro-Mechanical, Bearings, ...	Engineer.	Desktop
SCANeR	AD, ADAS, Roads, Vehicle dynamics, Traffi...	Engineer.	Desktop
Scenario-Runner	CARLA, OpenScenario	Engineer.	OSS (C)

Tool	Features	Use Case	Forma
SCILAB & Xcos	Numerical Computation Solver, Controls, ...	Engineer.	OSS (C
SILAB	Driving Simulator, Dynamics, HIL	Engineer.	Deskt
Simcenter 3D software	CAE, CAD, NVH, Vibration, Acoustics, Mec...	Engineer.	Deskt
Simcenter Amesim	CAE, Electrical, Fluid, Thermal	Engineer.	Deskt
Simcenter FLOEFD	CAE, CFD	Engineer.	Deskt
Simcenter MADYMO	Occupant & Pedestrian Safety, Crash, Car/...	Engineer.	Deskt
Simcenter Prescan	ADAS, AD, Sensors, Radar, Camera, MBD	Engineer.	Deskt
Simcenter SPH Flow	CAE, CFD, Smoothed-Particle Hydrodyna...	Engineer.	Deskt
Simcenter STAR-CCM+ CFD	CAE, CFD, Multiphysics	Engineer.	Deskt
Simcenter Tire	Tires	Engineer.	Deskt
Simpack (SIMULIA)	Multidomain, MBD, NVH, SIL, HIL, Vehicle ...	Engineer.	Deskt
Simplorer	Modelica, Multidomain, MBD, Dynamics, F...	Engineer.	Deskt
SimulationX - Multi-Body Syste...	Modelica, Multidomain, Multi-Body Syste...	Engineer.	Deskt
Simulink Vehicle Dynamics Blo...	Vehicle Dynamics, MATLAB/Simulink, MB...	Engineer.	Deskt
SolidWorks Simulation	CAE, FEA, CAD, Structures	Engineer.	Deskt
Stage (2D Multi-Robot Simulat...	Robotics, Multi-Agent, ROS	Engineer.	OSS (C
Stride (formerly Xenko)	Game Development, 2D, 3D, C#, Windows	Game	OSS (C
SUMO (Simulation of Urban M...	Traffic, Traffic-Manipulation, Maps, Road-...	Engineer.	OSS (C
SUMO-netconvert	Traffic, Road File Conversion	Engineer.	OSS (C
SuspensionSim	Suspension	Engineer.	Deskt
Tecplot 360	CAE, CFD	Engineer.	Deskt
TORCS (The Open Racing Car...	Driving Simulator, Dynamics	Game/Eng...	OSS (S
Torque3D	Game Development, 2D, 3D, C++ Engine	Game	OSS, C
Tosca	CAE, FEA, CFD, Optimization	Engineer.	Deskt
trafficSimulator	Traffic, Python	Engineer.	OSS (C
TransModeler	Traffic, Maps, 3D Visualization	Engineer.	Deskt
TSIS-CORSIM	Traffic, Maps	Engineer.	Deskt
Udacity	Deep Learning, AD, ADAS	Engineer.	OSS (C
Unity	Game Development, 3D, C++ Engine, Sen...	Game/Eng...	Deskt
Unreal Engine	Game Development, 3D, C++ Engine, Sen...	Game/Eng...	Deskt
VA ONE	NVH, Acoustics, Vibration	Engineer.	Deskt
Veins	Inter Vehicle Communication, SUMO, Traffic	Engineer.	OSS (C
VI-CarRealTime	Vehicle Dynamics, Design of Experiments,...	Engineer.	Deskt

Tool	Features	Use Case	Forma
VIRTUAL PERFORMANCE SOL...	Crash, Safety, Impact, Airbag, Battery, Str...	Engineer.	Deskt
Virtual Test Drive (VTD)	Environment, OpenDRIVE, OpenSCENARI...	Engineer.	Deskt
Vissim	Traffic, Maps, Digital City Twin, OpenStre...	Engineer.	Deskt
Webots	ROS Robotics Simulator	Engineer.	OSS (C
Wolfram SystemModeler	Modelica, Multidomain, MBD, Dynamics, F...	Engineer.	Deskt
XFlow CFD	CFD, Mesh-free, Lattice Boltzmann Appro...	Engineer.	Deskt

Disclaimer: Table is WIP!

Abbreviations

AD / ADAS: Autonomous Driving / Advanced driver-assistance systems

CAE / CAD: Computer Aided Engineering/Design (Physical modeling of structure)

CFD: Computational fluid dynamics

FEA / FEM: Finite Elements Analysis / Finite Element Method

MBD: Model-Based Development

HiL: Hardware in the Loop

SIL: Software in the Loop

This page tries to answer questions similar to:

- Which simulation tools can be used for AD, ADAS verification.
- What game engines can simulate cars?
- What driving simulators are there?
- Which tools can analyze dynamics of cars?

Would you like us to edit or add any entries to the table above? Please let us know over at "Contact".

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